# Counterfactual Training

## Teaching Models Plausible and Actionable Explanations

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# Dynamic HTML Version

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# Training Opaque Models

*Tweaking Parameters*

**Objective**:

$$\min_{\theta} \{\mathsf{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$

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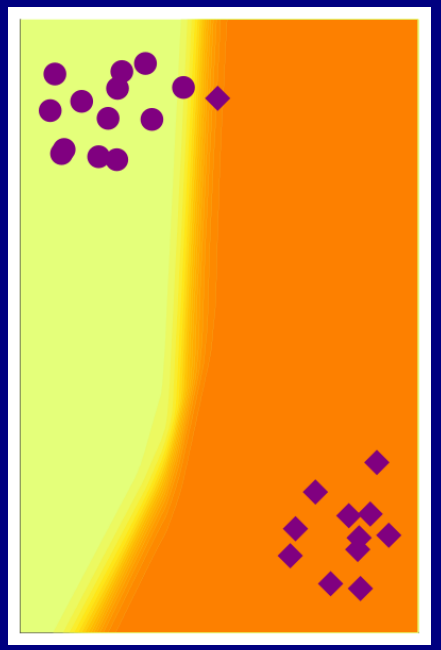
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**Objective**:

$$\min_{\theta}\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$

**Solution**:

$$\theta_{t+1} = \theta_t - \nabla_\theta\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$
$$\theta^* = \theta_T$$

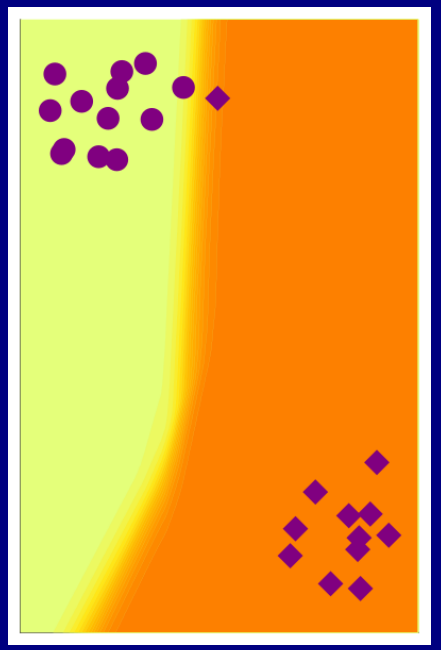


# Explaining Opaque Models

*Tweaking Inputs*

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$$\min_{\mathbf{x}} \{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda \text{reg}(\mathbf{x}; \cdot)\}$$



# Explaining Opaque Models

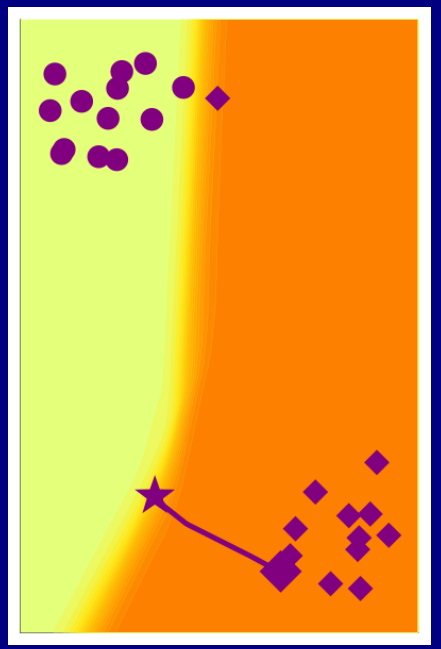
*Tweaking Inputs*

**Objective**:

$$\min_{\mathbf{x}} \{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda \text{reg}(\mathbf{x}; \cdot)\}$$

**Solution**:

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{x}_t - \nabla_{\mathbf{x}} \{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) \\ &\quad + \lambda \text{reg}(\mathbf{x}; \cdot)\} \\ \mathbf{x}^* &= \mathbf{x}_T \end{aligned}$$



# Explanation or Adversarial Example?

# Plausibility at all cost?

All of these counterfactuals are valid explanations for the model's prediction.

*Pick your poison ...*

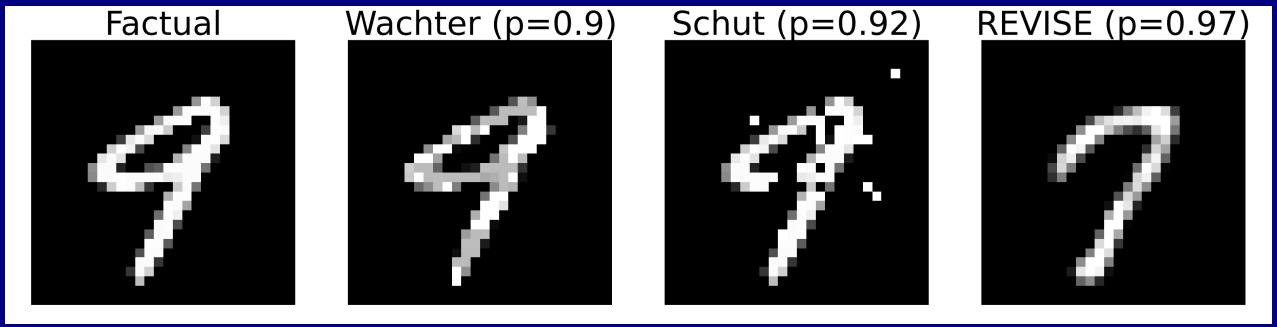



Figure 1: Turning a 9 into a 7: Counterfactual explanations for an image classifier using different approaches (Altmeyer et al. 2024).

# Faithful First, Plausible Second

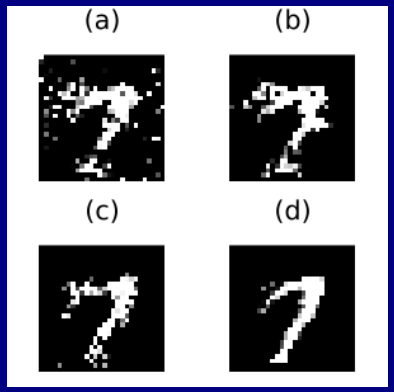



Figure 2: Turning a 9 into a 7. *ECCCo* applied to MLP (a), Ensemble (b), JEM (c), JEM Ensemble (d).

🔑 **Insight**: faithfulness facilitates[1]

- model quality checks (Figure 2).
- state-of-the-art plausibility (Figure 3).

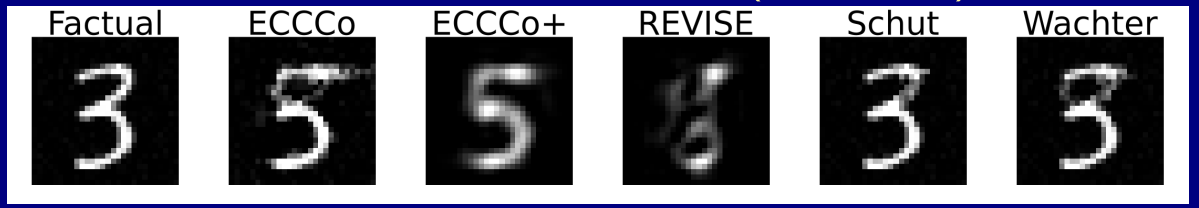



Figure 3: Results for different generators (from 3 to 5).

[1] Altmeyer et al. (2024) @ AAAI 2024. [blog]

# Putting it all together

# Counterfactual Training

First, *Tweaking Inputs*[2]

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \nabla_{\mathbf{x}}\{ECCCo(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+)\}$$
$$\mathbf{x}^* = \mathbf{x}_T$$

Then, *Tweaking Parameters*

$$\theta_{t+1} = \theta_t - \nabla_{\theta}\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y}) + \text{div}(\mathbf{x}^*, \mathbf{x}^+, y^+; \theta)\}$$
$$\theta^* = \theta_T$$

[2] Generate faithful explanations using *ECCCo* objective (Altmeyer et al. 2024).

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3. Bonus: use nascent CE as AE → **Robustness** ↑

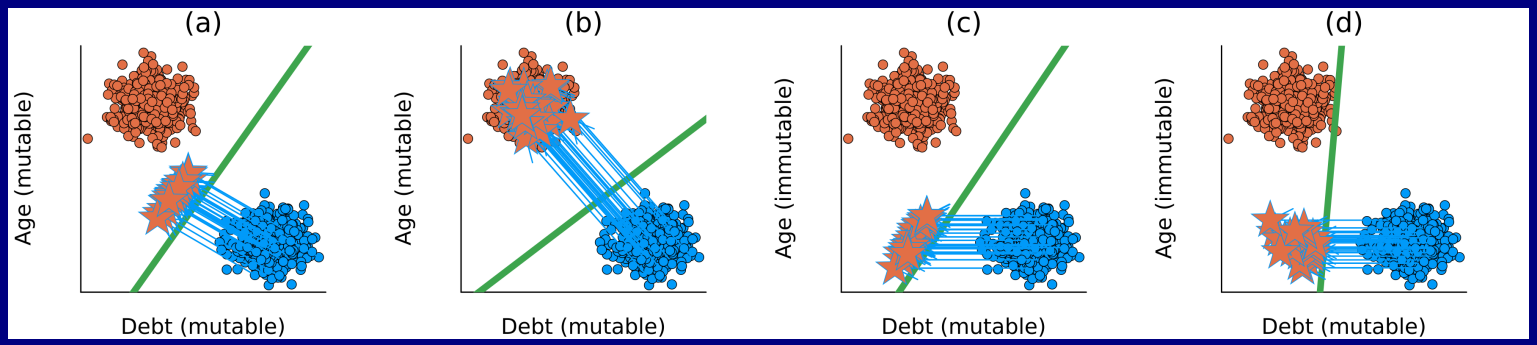



Figure 4: (a) conventional training, all mutable; (b) CT, all mutable; (c) conventional, *age* immutable; (d) CT, *age* immutable.

# Counterfactual Training: Results



Figure 5: **Plausibility**: Visual explanations (counterfactuals) for baseline (top row) vs CT (bottom).

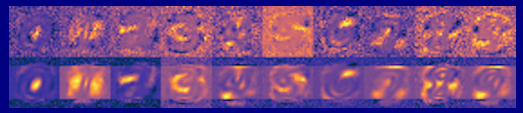


Figure 6: **Actionability**: Visual explanations (integrated gradients) as before. Five top and bottom rows immutable.

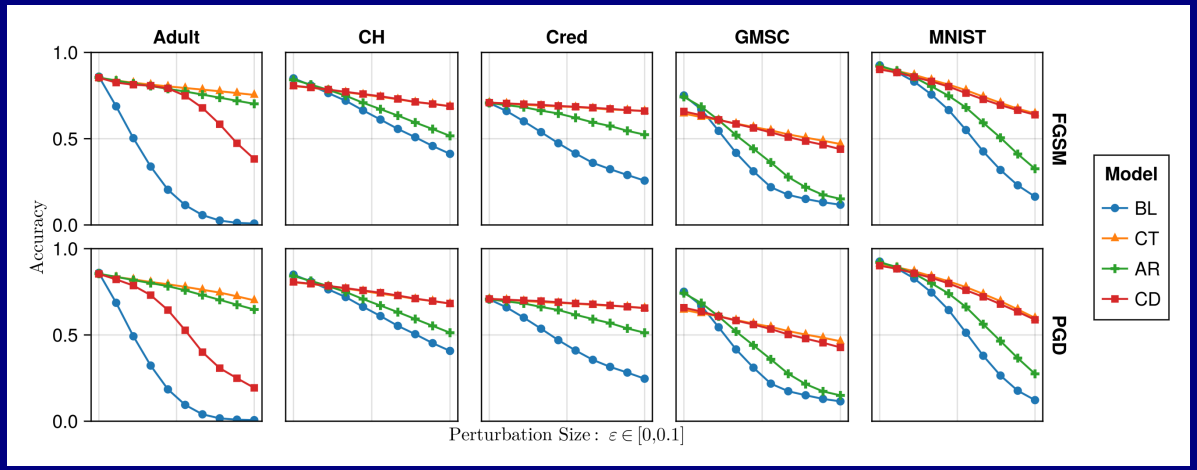



Figure 7: Test accuracies on adversarially perturbed data with varying perturbation sizes.

# The Hard Numbers

Extensive experiments and ablation studies on nine datasets–synthetic, tabular and vision–generating millions of counterfactuals:[3]

1. **Plausibility** of CEs increases by up to 90%.
2. **Actionability**: cost of reaching valid counterfactuals with protected features decreases by 19% on average.
3. Models' adversarial **robustness** improves consistently.

[3] Facilitated by our CounterfactualExplanations.jl (Altmeyer et al. 2023) with multi-processing support and (DHPC) (2022).

# Check it out!



Figure 8: Preprint



Figure 9: Software




Figure 10: Homepage

# References

Altmeyer, Patrick, Arie van Deursen, and Cynthia C. S. Liem. 2023. "Explaining Black-Box Models through Counterfactuals." *Proceedings of the JuliaCon Conferences* 1: 130. https://doi.org/10.21105/jcon.00130.

Altmeyer, Patrick, Mojtaba Farmanbar, Arie van Deursen, and Cynthia C. S. Liem. 2024. "Faithful Model Explanations through Energy-Constrained Conformal Counterfactuals." *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence* 38: 10829–37. https://doi.org/10.1609/aaai.v38i10.28956.

(DHPC), Delft High Performance Computing Centre. 2022. *DelftBlue Supercomputer (Phase 1)*. Https://www.tudelft.nl/dhpc/ark:/44463/DelftBluePhase1.